The empirical recurrence for OEIS A250450: recovering a conjecture that is not written down

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Abstract

OEIS A250450 records "Empirical recurrence of order 50", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 52$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 50, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 50 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250450 is "Number of (n+1)X(4+1) 0..2 arrays with nondecreasing min(x(i,j),x(i,j-1)) in the i direction and nondecreasing min(x(i,j),x(i-1,j)) in the j direction.". It has offset 1 and begins

$$4535, 141857, 3865967, 104578590, 2750712613, 71984404630, 1868402502223, 48446319776685, \ldots$$

The entry states:

> Empirical recurrence of order 50 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 12:25:52, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 243$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 52$ of the 243, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 52$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 104$ exact terms of the model returns order 50 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{50} c_i\, a(n-i),$$

c_1	22	c_{14}	7876869770343	c_{27}	−441383527646440448	c_{40}	−228499274294886400
c_2	488	c_{15}	−30798221962040	c_{28}	−824151167412542976	c_{41}	349684555281596416
c_3	−9132	c_{16}	−101075417781240	c_{29}	933941591540119552	c_{42}	44676738902392832
c_4	−80306	c_{17}	292630298951028	c_{30}	1415422001832554496	c_{43}	−98569551335129088
c_5	1357702	c_{18}	934037274114716	c_{31}	−1554430099503511552	c_{44}	−1748246942711808
c_6	7165223	c_{19}	−2085641921057280	c_{32}	−1845691025422407680	c_{45}	18064170201120768
c_7	−102421004	c_{20}	−6352631617163136	c_{33}	2016958316145991680	c_{46}	−1252383472484352
c_8	−414321073	c_{21}	11420587093756400	c_{34}	1806395887186329600	c_{47}	−1912973064929280
c_9	4521378634	c_{22}	32337794397951328	c_{35}	−2015892646373261312	c_{48}	263354509688832
c_{10}	16219820510	c_{23}	−48864268543829312	c_{36}	−1301134172820733952	c_{49}	87668872445952
c_{11}	−126241831040	c_{24}	−124661056000256128	c_{37}	1527834536109015040	c_{50}	−16698832846848
c_{12}	−431238134132	c_{25}	165002974160400768	c_{38}	667056981069791232		
c_{13}	2356989176038	c_{26}	366513475616770432	c_{39}	−860119758260928512		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 50, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $50+4$ terms removed returns order 50 as well, so $\deg q_{\min} = 50 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $50+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 50 that this sequence satisfies — and that family turns out to have exactly one member.

References

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